

1. (Discrete Random Variables)

Given the probability mass function:

$$P(X=x) = \begin{cases} k(2x+1), & x=1,2,3,4 \\ 0, & \text{otherwise} \end{cases}$$

a.) Value of the constant k

Since all probabilities must sum to 1:

$$k[(2(1)+1) + (2(2)+1) + (2(3)+1) + (2(4)+1)] = 1$$

$$k(3+5+7+9) = 1 \Rightarrow k(24) = 1 \Rightarrow k = \frac{1}{24}$$

x	1	2	3	4
P(X=x)	3/24	5/24	7/24	9/24

k ≈ 0.0417

b.) Compute E[X] and Var(X)

$$E[X] = \sum x \cdot P(x) = 1\left(\frac{3}{24}\right) + 2\left(\frac{5}{24}\right) + 3\left(\frac{7}{24}\right) + 4\left(\frac{9}{24}\right) = \frac{3+10+21+36}{24} = \frac{70}{24} \approx 2.9167$$

$$E[X^2] = \sum x^2 \cdot P(x) = 1^2\left(\frac{3}{24}\right) + 2^2\left(\frac{5}{24}\right) + 3^2\left(\frac{7}{24}\right) + 4^2\left(\frac{9}{24}\right) = \frac{3+20+63+144}{24} = \frac{230}{24} \approx 9.5833$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{230}{24} - \left(\frac{70}{24}\right)^2 = \frac{155}{144} \approx 1.0764$$

c. Calculate P(X ≥ 3)

$$P(X \geq 3) = P(X=3) + P(X=4) = \frac{7}{24} + \frac{9}{24} = \frac{16}{24} = \frac{2}{3} \approx 0.6667$$

2. (Binomial Distribution)

Given: n = 20, p = 0.08, q = 1 - p = 0.92 PMF formula: $P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$

$$a.) P(X=2) = \binom{20}{2} (0.08)^2 (0.92)^{18} = 190 \cdot 0.0064 \cdot 0.222936 \approx 0.2711$$

$$b.) P(X \geq 3) = 1 - P(X < 3) = 1 - [P(X=0) + P(X=1) + P(X=2)]$$

$$P(X=0) = \binom{20}{0} (0.08)^0 (0.92)^{20} \approx 0.1887 \rightarrow P(X \geq 3) = 1 - [P(0) + P(1) + P(2)] = 1 - 0.7879 \approx 0.2121$$

$$P(X=1) = \binom{20}{1} (0.08)^1 (0.92)^{19} \approx 0.3282$$

$$P(X=2) \approx 0.2711$$

c. Mean & SD

$$\text{Mean: } \mu = np = 20(0.08) = 1.6000$$

$$\text{SD: } \sigma = \sqrt{np(1-p)} = \sqrt{20(0.08)(0.92)} = \sqrt{1.472} \approx 1.2133$$

3. (Poisson Distribution)

$\lambda = 4.5$ hits per minute PMF formula: $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$

$$a.) P(X=6) = \frac{e^{-4.5} (4.5)^6}{6!} = \frac{(0.0111090)(8303.7656)}{720} \approx 0.1281$$

$$b.) P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$P(0) = \frac{e^{-4.5} (4.5)^0}{0!} \approx 0.0111 \quad P(1) = \frac{e^{-4.5} (4.5)^1}{1!} \approx 0.0500 \quad P(2) = \frac{e^{-4.5} (4.5)^2}{2!} \approx 0.1125$$

$$P(X \leq 2) = 0.0111 + 0.0500 + 0.1125 = 0.1736$$

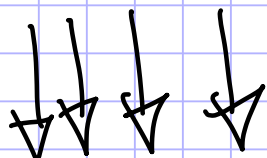
c.) P(X > 10) over a 2-minute interval

For t = 2 mins, adjusted rate $\lambda_2 = 4.5 \cdot 2 = 9$, so $X \sim \text{Poisson}(9)$

$$P(X > 10) = 1 - P(X \leq 10) = 1 - \sum_{x=0}^{10} \frac{e^{-9} (9)^x}{x!} \quad \lambda = 9: P(X \leq 10) \approx 0.7060$$

$$P(X > 10) = 1 - 0.7060 = 0.2940$$

Problem #4



4. (Continuous Probability & Normal Distribution)

Given: $\mu = 120 \text{ ms}$, $\sigma = 15 \text{ ms}$ a.) $P(X > 135)$

$$Z = \frac{135 - 120}{15} = 1.00 \quad P(X > 135) = P(Z > 1.00) = 1 - \Phi(1.00) = 1 - 0.8413 = \boxed{0.1587}$$

b.) $P(100 < X < 130)$

$$X_1 = 100: Z_1 = \frac{100 - 120}{15} = \frac{-20}{15} \approx -1.3333 \quad X_2 = 130: Z_2 = \frac{130 - 120}{15} = \frac{10}{15} \approx 0.6667$$

$$P(100 < X < 130) = P(-1.3333 < Z < 0.6667) = \Phi(0.6667) - \Phi(-1.3333)$$

$$\Phi(0.6667) \approx 0.7475$$

$$\Phi(-1.3333) \approx 0.0912$$

$$P(100 < X < 130) = 0.7475 - 0.0912 \approx \boxed{0.6563}$$

c.) 95th percentileFind z s.t. $P(Z < z) = 0.95$.From standard normal distribution tables, $z_{0.95} \approx 1.6449$.

$$X_{0.95} = \mu + z\sigma = 120 + 1.6449(15) = 120 + 24.6735 \approx \boxed{144.6735 \text{ ms}}$$